Host genomic susceptibility to severe respiratory syncytial virus infection

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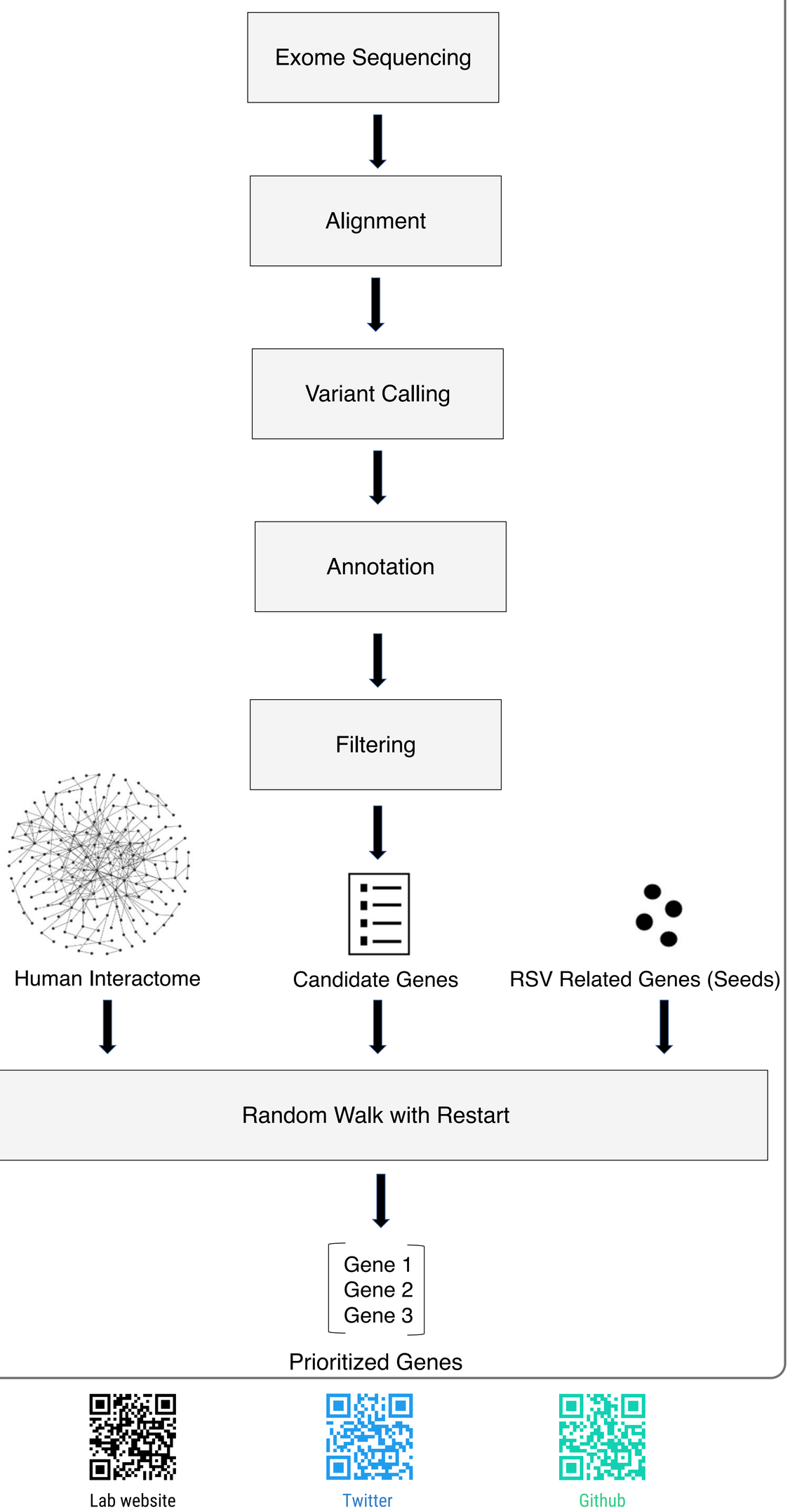
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Background

- Infection with **respiratory syncytial virus (RSV)** is mild in most children, still a fraction suffers from severe disease.
- Life-threatening susceptibility to RSV infection in previously healthy children can be indicative of pathogen-specific **primary immunodeficiencies**.
- Linking host-genetic factors to severe manifestation of RSV infection have implications for **prevention and treatment** of susceptible children.

Aim of the study: to find inborn errors of immunity predisposing previously healthy children to severe clinical presentation of RSV infection.

Project overview



Lab website | Twitter | Github

Data and methods

Data Whole-exome of 127 previously healthy children who needed hospitalization and oxygen therapy due to RSV infection.

Filtering For finding candidate variants, a Bayesian framework of ACMG/AMP guidelines was implemented. In summary, we gather all the evidences of pathogenicity and benignity for each variant, then an odds-of-pathogenicity is assigned to each variant. Variants with odds-of-pathogenicity >0.8 was considered as potentially damaging.

Statistical method We used Random Walk with Restart (RWR) to prioritize damaging variants. This method needs three inputs:

1. candidate genes (output of filtering step)
2. protein-protein interaction (PPI) network, here we used the STRING interactome.
3. disease related genes known as ”seeds”. in this study, seeds are RSV-associated genes/proteins from DisGeNET and OpenTargets.

RWR explores the vicinity of seeds to study their function, assuming that nodes with similar function tend to lie close to each other in the interactome.

To control for the impact of network structure on RWR output, we repeated the process with 1000 random seed sets, and calculated the p-value for each gene. Genes with false-discovery-rate <5% were kept.

Results

We discovered 28 variants in the genes of interest, including:

- Three ***IFIH1*** loss-of-function. This gene encodes an RIG-I-like receptor involved in the sensing of viral RNA. It is previously shown that the mutated *IFIH1* results in proteins which are unable to induce IFN-β, are intrinsically less stable, and lack ATPase activity.
- A frame-shift ***NOD2*** variant resulting a lack of synergistic response to RSV stimulation.
- A stop-gained ***IFI44L*** variant known as a risk factor for multisystem inflammatory syndrome in children.

Conclusions

- **Children carrying specific variants in immune-related genes are more susceptible to severe RSV infection.** This study can help to identify potential novel targets for diagnostics or therapeutic development.
- **This study provides further insights into the human genetic role against infection**, which could lead to a better understanding of interplay between host and pathogen.
- **This study identifies monogenic inborn errors of immunity;** however, a considerable number of infected children have no identified genetic defect yet. Further studies with a larger sample size is required to detect a collection of genetic variants affecting the key pathways in response to viral infection.

References

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The authors declare that they have no conflict of interest.